

Where Have All the Lemons Gone? Signaling and Cosigning in the U.S. Mortgage Market

Marius Grünewald¹

May 26, 2026

¹European University Institute and European Central Bank

Motivation (US)

- Equity is the main financial asset for many households; mortgages are the key lever to homeownership.
- In 2023, about 92% of new home purchases in the US used a mortgage.
- The US mortgage market is \$12.59 trillion ($\approx 70\%$ of consumer debt; $\approx 50\%$ of GDP) (2023).

Motivation (US)

- Equity is the main financial asset for many households; mortgages are the key lever to homeownership.
- In 2023, about 92% of new home purchases in the US used a mortgage.
- The US mortgage market is \$12.59 trillion ($\approx 70\%$ of consumer debt; $\approx 50\%$ of GDP) (2023).

Definition: What is a cosigner?

- A cosigner is not the main borrower, but is **legally responsible for repayment if the borrower defaults**.
- Cosigners **do not obtain ownership rights** in the property.

Table 1: Global Cosigning and Guarantee Shares

Country	Share	Primary Mechanism
United States	5.0%	Private (Cosigning)
Portugal	21.0%	Private (<i>Fiadores</i>)
Italy	20.0%	Public (<i>Consap Fund</i>)
Canada	13.0%	Private (Co-borrowing)
Spain	15.0%	Private (<i>Aval</i>)
Australia	10.0%	Private (Guarantor)
Norway	15.0%	Private (Co-borrower)
Finland	19.0%	Public (ASP)
United Kingdom	7.0% (+1.4%)	Private (JBSP), Government Guarantee Scheme

Mechanism and Research Question

Research question

- Why is the share of cosigned mortgages in the US so small?

Mechanism and Research Question

Research question

- Why is the share of cosigned mortgages in the US so small?

Key mechanism

- Interpret cosigning as insurance and signal.
- Institutional decoupling of acceptance and pricing creates adverse selection.
- Signal partially offsets adverse selection.

Mechanism and Research Question

Research question

- Why is the share of cosigned mortgages in the US so small?

Key mechanism

- Interpret cosigning as insurance and signal.
- Institutional decoupling of acceptance and pricing creates adverse selection.
- Signal partially offsets adverse selection.

Why should we care?

- **Housing affordability:** First order welfare effects from housing access.
- **Financial stability:** Will it introduce moral hazard?

Preview of Results

Empirics:

1. Intergenerationally cosigned mortgages account for about 5% of mortgages in the US (\$12.3 billion p.a.).
2. Cosigned applicants are younger, live in pricier areas and have higher housing demand.
3. Natural Experiment: Cosigners are regulated on the intensive, solo applicants on the extensive margin.

Theory:

1. Insurance property of cosigning attracts bad applicants.
2. Adverse selection and signaling create three regions:

Quantitative Counterfactual exercise: Joint Pricing instead of Lower Median Pricing

1. 20% reduction of cross country difference.
2. Increased access to mortgages and homeownership 1%.
3. No increased default rates.

1. Literature
2. Data
3. Descriptives
4. Theoretical Model
5. Natural Experiment
6. Quantitative Model

Literature

- **Housing Finance and mortgage markets:** Bhutta-Hizmo-Ringo (2025, JF), Adelino-Schoar-Severino (2016, RFS), Hurst-Keys-Seru-Vavra (2016, AER), DeFusco-Paciorek (2026), Kaplan-Mitman-Violante (2020, JPE), DeFusco-Tang-Yannelis (2022, JFinEcon)

Documenting cosigning and GSE pricing schedules.

- **Parental Help for Homeownership:** Allen-Hardin-Sirmans (2026, JME), Benetton-Eckerd-Ramchara (2022), Engelhardt-Mayer (1998, JUE), Guiso-Jappelli (2002, JMBC), Cox (1987, JPE), Boar (2021, ReStud), Barczyk-Kredler (2018, ReStud), Daruich (2020, RED), Dreudahl-Martinell (2022, ReStat),

Parental help as signal and insurance.

- **Adverse Selection and Signaling in Credit Markets:** Stiglitz-Weiss (1981, AER), Rothschild-Stiglitz (1976, QJE), Spence (1973, QJE), Akerlof (1970, QJE), Bester (1985, AER), Karlan-Zinman (2009, ECMA), Adams-Einav-Levin (2009, AER), Einav-Finkelstein-Cullen (2010, QJE), Crawford-Pavanini-Schivardi (2018, AER), Stroebel (2016, JPE), Agarwal-Ben-David-Yao (2017, JFinEcon), Holmström-Tirole (1997, QJE)

Third-Party Signal and Adverse Selection in Credit Markets.

Data

Table 2: Key Features of HMDA, Web-Scraped Dat

Feature	HMDA
Scope	Near-universe of loan-level applications (2018–2024)
Key Info	Mortgage, Borrower, Lender details; Application outcomes (originated/rejected)
Cosigning ID	Via age structure ($\geq$ 19-year age difference)
Drawback	Only first two mortgagors listed

Table 2: Key Features of HMDA, Web-Scraped Dat

Feature	HMDA	Web-Scraped Data
Scope	Near-universe of loan-level applications (2018–2024)	Universe of originated mortgages in a given county (currently 30)
Key Info	Mortgage, Borrower, Lender details; Application outcomes (originated/rejected)	Mortgagors, Mortgagees, Loan Amount, Geographic Area, Origination Date
Cosigning ID	Via age structure ($\geq$ 19-year age difference)	Via more than two mortgagors listed
Drawback	Only first two mortgagors listed	No age information

Descriptives

HMDA - Evidence on Cosigning

2.6% of mortgage applications are intergenerationally cosigned. (7.4 billion USD p.a.)

With webscraped data: 5.47%

Table 3: By Age

Age Bin	Cosigned (%)	w/ Webscraping
<25	9.57	12.36
25–34	2.99	5.60
35–44	1.30	4.04

HMDA - Evidence on Cosigning

2.6% of mortgage applications are intergenerationally cosigned. (7.4 billion USD p.a.)

With webscraped data: 5.47%

Table 3: By Age

Age Bin	Cosigned (%)	w/ Webscraping
<25	9.57	12.36
25–34	2.99	5.60
35–44	1.30	4.04

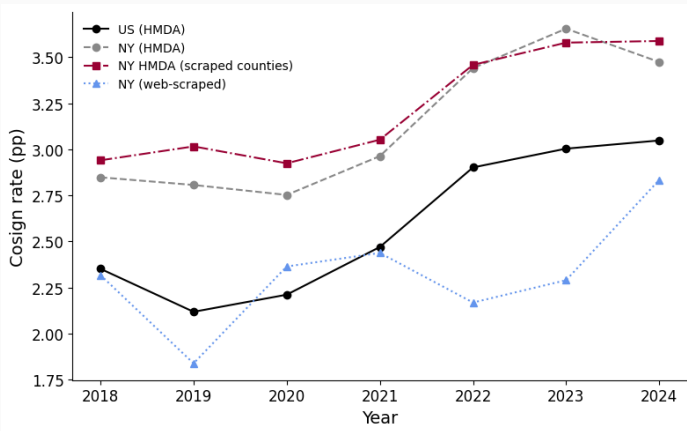
Table 4: By Year

Year	Cosigned (%)	w/ Webscraping
2018	2.55	5.41
2019	2.12	4.37
2020	2.21	5.04
2021	2.41	5.16
2022	2.88	5.55
2023	3.11	5.64
2024	3.19	7.01

States

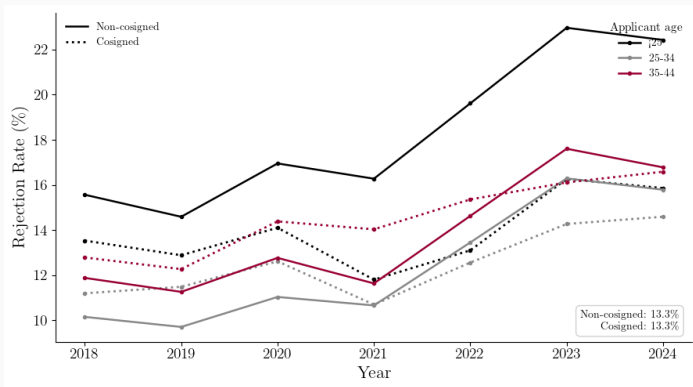
How does webscraping compare?

Figure 1: Webscraped sample in comparison to HMDA



Fact 1 - Application Rejection Rates

Figure 2: Mean Difference in Application Rejection Rates

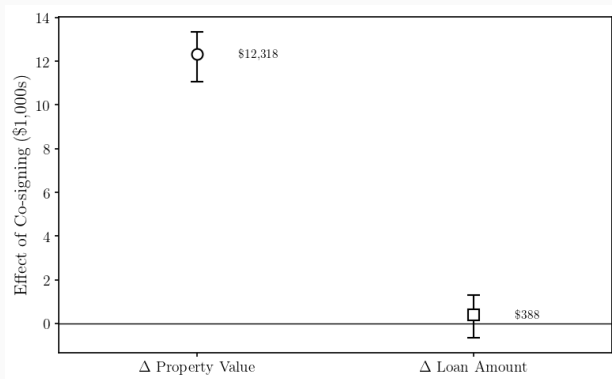


Unweighted Average across Property Value x Loan Amount cell.

Fact 2 - Housing Demand

Conditional on Acceptance

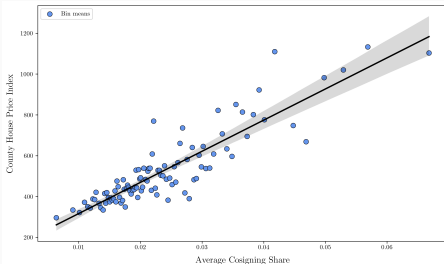
Figure 3: Mean Difference in Property Value and Loan Amount



Fact 3 - House Prices

Fact 3.: Cosigned Mortgagees live in pricier areas

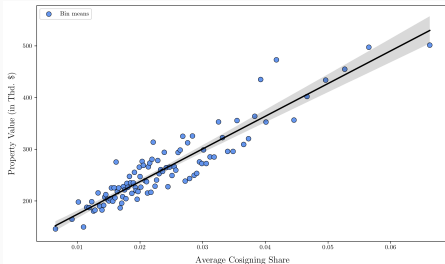
County House Price Index



House Price Index

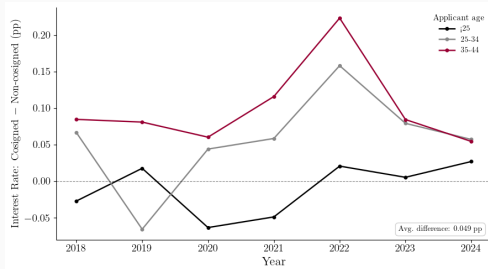
Larger Counties

Avg. Property Value



Fact 4 - Interest Rate

Figure 4: Mean Difference in Interest Rates



Weighted Average across Property Value x Loan Amount cell.

► w/ Web scraping

Theoretical Model

The model in a nutshell

Cosigning does two things

- Insurance in default.
- A signal of creditworthiness to the bank.

The institutional friction

- Banks first **admit** application, then **price** the mortgage.
- Joint income eases admission but lower income pricing.

The model in a nutshell

Cosigning does two things

- Insurance in default.
- A signal of creditworthiness to the bank.

The institutional friction

- Banks first **admit** application, then **price** the mortgage.
- Joint income eases admission but lower income pricing.

Consequence: skewed pool quality

- Lower entry on observables is offset by the cosigner's income, so more applications are admitted.
- **Riskier borrowers** sort disproportionately into cosigned mortgages.
- The pool of approved cosigned loans is therefore **adversely selected**.

The model in a nutshell

Cosigning does two things

- Insurance in default.
- A signal of creditworthiness to the bank.

The institutional friction

- Banks first **admit** application, then **price** the mortgage.
- Joint income eases admission but lower income pricing.

Consequence: skewed pool quality

- Lower entry on observables is offset by the cosigner's income, so more applications are admitted.
- **Riskier borrowers** sort disproportionately into cosigned mortgages.
- The pool of approved cosigned loans is therefore **adversely selected**.

The signal gets diluted

- Banks internalize the adverse selection when pricing.
- The interest-rate discount on cosigned mortgages is **dampened**: the positive guarantee signal is offset by the worse underlying borrower pool.

Three Players, One Friction

Young Borrower

- Observable income y
- Private type $\theta \in \{G, B\}$
 $\pi_G > \pi_B$
(repayment prob.)
- Richer $\Rightarrow$ more likely Good:
 $\lambda(y) = \Pr(\theta=G \mid y) \uparrow$
- Chooses mortgage menu j

Parent

- **Observes** child's type θ
- Stronger credit:
 $\lambda(y_p) > \lambda(y)$
- Altruism β , liability cost Δu_p
- Decides whether to guarantee
 $z \in \{0, 1\}$

Bank

- Perfectly competitive
- Observes y and signal z
- **Does not** observe θ
- Posts zero-profit rate $r_j(y)$

Two Mortgage Products

Solo mortgage

- Qualifying income: $Y_{so} = y$
- DTI cap κ may **bind** $\Rightarrow$ rejection
- Loan: $L_{so} \leq \kappa y / (1 + r_{so})$
- Default $\Rightarrow$ lose house *and* face cost ϕ :
 $\underline{u}_{so} = u(y - \phi, 0)$

Cosigned mortgage

- Qualifying income: $Y_{co} = y + y_p$
- DTI cap **does not bind**
- Loan capped institutionally at L^*
- Default $\Rightarrow$ parent covers; child keeps house:
 $\underline{u}_{co} = u(y - \phi, h_{co})$

Two Mortgage Products

Solo mortgage

- Qualifying income: $Y_{so} = y$
- DTI cap κ may **bind** $\Rightarrow$ rejection
- Loan: $L_{so} \leq \kappa y / (1 + r_{so})$
- Default $\Rightarrow$ lose house *and* face cost ϕ :
 $\underline{u}_{so} = u(y - \phi, 0)$

Cosigned mortgage

- Qualifying income: $Y_{co} = y + y_p$
- DTI cap **does not bind**
- Loan capped institutionally at L^*
- Default $\Rightarrow$ parent covers; child keeps house:
 $\underline{u}_{co} = u(y - \phi, h_{co})$

The key asymmetry in default:

$$\Delta \underline{u} \equiv \underline{u}_{co} - \underline{u}_{so} > 0 \quad \text{because } h_{co} > 0$$

Cosigning provides **two benefits**: unlocks a larger loan *and* insures the borrower in default.

The Engine: Bad Types Love Cosigning

The net gain from cosigning decomposes as:

$$\Delta V(y, \theta) = \underbrace{\pi_\theta \cdot G(y)}_{\text{repayment-state gain}} + \underbrace{(1 - \pi_\theta) \cdot \Delta \underline{u}}_{\text{default-state insurance}}$$

Good type (high π_G)

- Rarely defaults $\Rightarrow$ weights $G(y)$ heavily
- Will switch to solo when solo gets good enough

Bad type (low π_B)

- Defaults often $\Rightarrow$ weights $\Delta \underline{u}$ heavily
- *Clings* to cosigning longer

The Engine: Bad Types Love Cosigning

The net gain from cosigning decomposes as:

$$\Delta V(y, \theta) = \underbrace{\pi_\theta \cdot G(y)}_{\text{repayment-state gain}} + \underbrace{(1 - \pi_\theta) \cdot \Delta u}_{\text{default-state insurance}}$$

Good type (high π_G)

- Rarely defaults $\Rightarrow$ weights $G(y)$ heavily
- Will switch to solo when solo gets good enough

Bad type (low π_B)

- Defaults often $\Rightarrow$ weights Δu heavily
- *Clings* to cosigning longer

Adverse Selection Pressure

Bad types have a *stronger* preference for the cosigned contract than Good types.

$\Rightarrow$ Any cosign pool attracts disproportionately Bad types.

Who Moves When

1. **Nature** draws the full state $(\theta, y, y_p, L, \eta_p)$.
Household observes everything; bank observes only y .
2. **Parent signals:** chooses guarantee $z \in \{0, 1\}$ to maximise altruistic payoff net of liability cost and idiosyncratic shock η_p .
3. **Bank prices:** observes (y, z) ; if $z=1$, also sees y_p .
Updates beliefs via Bayes' rule; posts zero-profit rate $r_j(y)$.
4. **Borrower sorts:** picks menu $j \in \{so, co\}$ (cosign feasible only if $z=1$)
to maximise expected utility $V_j(y, \theta)$.

Is $z=1$ a Credible Signal?

Parent cosigns iff benefit exceeds cost:

$$\beta \Delta V(y, \theta) - (1 - \pi_\theta) \Delta u_p \geq \eta_p$$

The tension: Bad types value cosigning *more* ($\Delta V_B > \Delta V_G$), so their parents have a *stronger* altruistic motive to cosign.

Is $z=1$ a Credible Signal?

Parent cosigns iff benefit exceeds cost:

$$\beta \Delta V(y, \theta) - (1 - \pi_\theta) \Delta u_p \geq \eta_p$$

The tension: Bad types value cosigning *more* ($\Delta V_B > \Delta V_G$), so their parents have a *stronger* altruistic motive to cosign.

What keeps the signal honest? The financial liability: Bad-type parents bear higher expected cost $(1 - \pi_B) > (1 - \pi_G)$.

Assumption: Cost Dominance

The financial penalty of covering a Bad kid's default exceeds the extra altruistic benefit of giving them a better mortgage.

$\Rightarrow q_G(y, y_p) > q_B(y, y_p)$: Good-type parents cosign more.

Is $z=1$ a Credible Signal?

Parent cosigns iff benefit exceeds cost:

$$\beta \Delta V(y, \theta) - (1 - \pi_\theta) \Delta u_p \geq \eta_p$$

The tension: Bad types value cosigning *more* ($\Delta V_B > \Delta V_G$), so their parents have a *stronger* altruistic motive to cosign.

What keeps the signal honest? The financial liability: Bad-type parents bear higher expected cost $(1 - \pi_B) > (1 - \pi_G)$.

Assumption: Cost Dominance

The financial penalty of covering a Bad kid's default exceeds the extra altruistic benefit of giving them a better mortgage.

$\Rightarrow q_G(y, y_p) > q_B(y, y_p)$: Good-type parents cosign more.

Bayesian update:

$$\Lambda_1(y) = \Pr(\theta=G \mid y, z=1) > \lambda(y) > \Lambda_0(y)$$

Seeing the cosign guarantee raises the bank's assessment of pool quality.

Institutional Friction Triggers the Exit of Good Types

Recall the net gain from cosigning:

$$\Delta V(y, \theta) = \underbrace{\pi_{\theta} \cdot G(y)}_{\substack{\text{repayment-state gain} \\ \uparrow \text{G-types weight this}}} + \underbrace{(1 - \pi_{\theta}) \cdot \Delta u}_{\substack{\text{default-state insurance} \\ \uparrow \text{B-types weight this}}}$$

(1) Wealth Effect

- Solo loan $L_{so}(y) \propto y$ **grows** with income
- Cosign loan $L_{co} = L^*$ **capped** by LTV
- $\Rightarrow$ Repayment-state gain $G(y) \downarrow$ as $y \uparrow$

(2) Rising Cosign Rate

- Entry via $Y_{co} = y + y_p$, pricing at solo risk
- B-types flood in $\Rightarrow$ pool quality $\mu(y) \downarrow$
- $r_{co}(y) = \frac{1+\rho}{\mu(y)} - 1$ **rises**, eroding $G(y)$ further

Institutional Friction Triggers the Exit of Good Types

Recall the net gain from cosigning:

$$\Delta V(y, \theta) = \underbrace{\pi_{\theta} \cdot G(y)}_{\substack{\text{repayment-state gain} \\ \uparrow \text{G-types weight this}}} + \underbrace{(1 - \pi_{\theta}) \cdot \Delta \underline{u}}_{\substack{\text{default-state insurance} \\ \uparrow \text{B-types weight this}}}$$

(1) Wealth Effect

- Solo loan $L_{so}(y) \propto y$ **grows** with income
- Cosign loan $L_{co} = L^*$ **capped** by LTV
- $\Rightarrow$ Repayment-state gain $G(y) \downarrow$ as $y \uparrow$

(2) Rising Cosign Rate

- Entry via $Y_{co} = y + y_p$, pricing at solo risk
- B-types flood in $\Rightarrow$ pool quality $\mu(y) \downarrow$
- $r_{co}(y) = \frac{1+\rho}{\mu(y)} - 1$ **rises**, eroding $G(y)$ further

G-types weight $G(y)$ heavily (π_G high); B-types weight $\Delta \underline{u}$ heavily (π_B low).

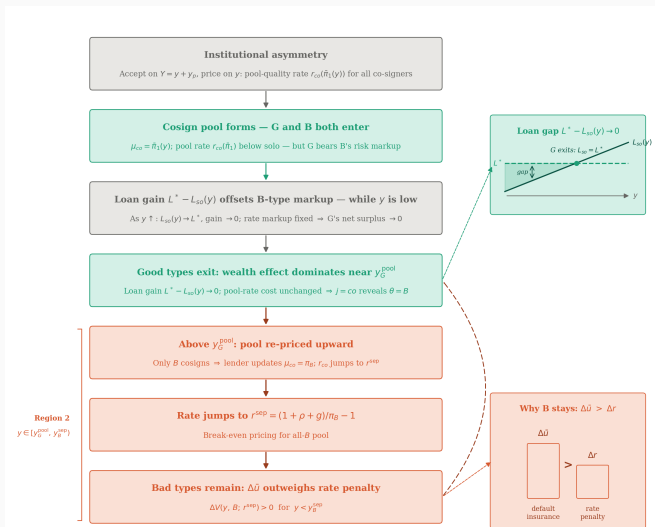
$\Rightarrow$ As $G(y)$ is squeezed, **Good types exit first**. Bad types stay for the insurance.

Lemma: Wealth Effect Dominates

Both forces compress $G(y) \Rightarrow \tilde{y}_G < \tilde{y}_B$: Good types exit at a *strictly lower* income than Bad types.

This is the entry point of adverse selection $\Rightarrow$ Region 2.

Figure 5: Adverse Selection Cycle



Proposition 1: The Three-Region Equilibrium

Two cutoffs partition the income distribution: $y_F < y_G^{pool} < y_B^{sep}$

Region	Who cosigns?	Bank's belief / rate
1: $y < y_G^{pool}$	Both Good & Bad types	Semi-Separating: $r_{co}^{pool}(y) = \frac{1+\rho+g}{\bar{\pi}_1(y)} - 1$
2: $y_G^{pool} \leq y < y_B^{sep}$	Bad types only	Separating: $r_{co}^{sep} = \frac{1+\rho+g}{\pi_B} - 1$
3: $y \geq y_B^{sep}$	Nobody	Cosign menu infeasible

Proposition 1: The Three-Region Equilibrium

Two cutoffs partition the income distribution: $y_F < y_G^{pool} < y_B^{sep}$

Region	Who cosigns?	Bank's belief / rate
1: $y < y_G^{pool}$	Both Good & Bad types	Semi-Separating: $r_{co}^{pool}(y) = \frac{1+\rho+g}{\bar{\pi}_1(y)} - 1$
2: $y_G^{pool} \leq y < y_B^{sep}$	Bad types only	Separating: $r_{co}^{sep} = \frac{1+\rho+g}{\pi_B} - 1$
3: $y \geq y_B^{sep}$	Nobody	Cosign menu infeasible

Region 1: Pooling equilibrium

Both type cosign but signal still informative.

Proposition 1: The Three-Region Equilibrium

Two cutoffs partition the income distribution: $y_F < y_G^{pool} < y_B^{sep}$

Region	Who cosigns?	Bank's belief / rate
1: $y < y_G^{pool}$	Both Good & Bad types	Semi-Separating: $r_{co}^{pool}(y) = \frac{1+\rho+g}{\bar{\pi}_1(y)} - 1$
2: $y_G^{pool} \leq y < y_B^{sep}$	Bad types only	Separating: $r_{co}^{sep} = \frac{1+\rho+g}{\pi_B} - 1$
3: $y \geq y_B^{sep}$	Nobody	Cosign menu infeasible

Region 1: Pooling equilibrium

Both type cosign but signal still informative.

Region 2: Adverse Selection Trap

Good types have left. Bad types remain. The bank knows this. The rate r_{co}^{sep} is the highest the market can sustain.

Prediction 1: Interest Rate Pass-Through

An increase in the guarantee fee g raises each borrower's rate by $1/\mu_j(y)$ where $\mu_j(y)$ is the income-dependent pool quality perceived by the bank for menu j :

Pool average quality:

- $\bar{\mu}_{co}$ mixes Region 1 and 2.
- $\bar{\mu}_{so}$ reflects the full income distribution
 $\Rightarrow \bar{\mu}_{co} < \bar{\mu}_{so}$ (adverse selection dominates)

Average rate pass-through: $\beta_{rate}^j \equiv \mathbb{E}[1/\mu_j(y) \mid j]$

$$\bar{\mu}_{co} < \bar{\mu}_{so} \implies \beta_{rate}^{co} > \beta_{rate}^{so}$$

Prediction 1: Interest Rate Pass-Through

An increase in the guarantee fee g raises each borrower's rate by $1/\mu_j(y)$ where $\mu_j(y)$ is the income-dependent pool quality perceived by the bank for menu j :

Pool average quality:

- $\bar{\mu}_{co}$ mixes Region 1 and 2.
- $\bar{\mu}_{so}$ reflects the full income distribution
 $\Rightarrow \bar{\mu}_{co} < \bar{\mu}_{so}$ (adverse selection dominates)

Average rate pass-through: $\beta_{rate}^j \equiv \mathbb{E}[1/\mu_j(y) \mid j]$

$$\bar{\mu}_{co} < \bar{\mu}_{so} \implies \beta_{rate}^{co} > \beta_{rate}^{so}$$

Proposition 2

When a common cost shock hits, cosigners should see a **larger** rate increase than non-cosigners.

Prediction 2: Rejection Insulation

Rejection occurs when $DTI_j = (1 + r_j)L/Y_j > \kappa$. The rejection pass-through is:

$$\beta_{rej}^j = \underbrace{\mathbb{E}[\sigma_j \mid DTI_j = \kappa]}_{\text{rate sensitivity at margin}} \times \underbrace{f_{DTI}^j(\kappa)}_{\text{density at margin}}$$

Two forces pull in opposite directions:

Rate sensitivity $\sigma_j = L/(Y_j \mu_j)$

$$\bar{\mu}_{co} < \bar{\mu}_{so} \Rightarrow \sigma_{co} > \sigma_{so}$$

Cosigners have higher DTI sensitivity.

Threshold density $f_{DTI}^j(\kappa)$

$$Y_{co} = y + y_p \gg y = Y_{so}$$

$\Rightarrow$ cosigners sit *far* from the cap

$$\Rightarrow f_{DTI}^{co}(\kappa) \ll f_{DTI}^{so}(\kappa)$$

Prediction 2: Rejection Insulation

Rejection occurs when $DTI_j = (1 + r_j)L/Y_j > \kappa$. The rejection pass-through is:

$$\beta_{rej}^j = \underbrace{\mathbb{E}[\sigma_j \mid DTI_j = \kappa]}_{\text{rate sensitivity at margin}} \times \underbrace{f_{DTI}^j(\kappa)}_{\text{density at margin}}$$

Two forces pull in opposite directions:

Rate sensitivity $\sigma_j = L/(Y_j \mu_j)$

$$\bar{\mu}_{co} < \bar{\mu}_{so} \Rightarrow \sigma_{co} > \sigma_{so}$$

Cosigners have higher DTI sensitivity.

Threshold density $f_{DTI}^j(\kappa)$

$$Y_{co} = y + y_p \gg y = Y_{so}$$

$\Rightarrow$ cosigners sit *far* from the cap

$$\Rightarrow f_{DTI}^{co}(\kappa) \ll f_{DTI}^{so}(\kappa)$$

When the **parental income buffer** is large enough, the sparsity channel dominates:

$$\beta_{rate}^{co} > \beta_{rate}^{so} \quad \text{yet} \quad \beta_{rej}^{co} < \beta_{rej}^{so}$$

Proposition 3

Despite larger rate pass-through, cosigners should be **insulated from rejection** relative to non-cosigners.

Natural Experiment

Primary mortgage market

- Households take out mortgages from banks, credit unions, and mortgage companies.
- Lenders screen and price loans.
- Rejections, interest rates, and cosigning are determined.

Primary mortgage market

- Households take out mortgages from banks, credit unions, and mortgage companies.
- Lenders screen and price loans.
- Rejections, interest rates, and cosigning are determined.

Secondary mortgage market

- Lenders sell eligible loans to Fannie Mae and Freddie Mac (GSEs).
- GSEs pool loans into MBS and guarantee investors against default.
- For this guarantee, GSEs charge guarantee fees (g-fees).

Reform in the Secondary Market

Reform (FHFA pricing change)

- Announced: January 5th, 2022. Implemented: April 1st, 2022.
- FHFA increased g-fees on **high-balance** loans (0.25 percent and 0.75 percent)
- Equal for Cosigned and Non-cosigned loans.
- Increase in marginal cost of banks for these loans.
- Interpretation: **Exogenous cost shock** at the conforming loan limit, which shifts banks' credit supply for loans just above the cutoff.

Exposure to GSE

Over time

Delinquency

Interest Rates

From Structural Pricing to Regression Estimand

Zero-profit pricing (log-linearized): $r^k \approx (\rho + g) - m^k$, $m^k \equiv \ln \mu^k$

Four cells:

Below

Pre $\rho + g_0 - m_{B,Pre}^k$

Post $\rho + \Delta\rho + g_0 - m_{B,Post}^k$

Above

$\rho + g_0 + \Delta g^{HB} - m_{A,Pre}^k$

$\rho + \Delta\rho + g_0 + \Delta g^{HB} + \Delta g^{reform} - m_{A,Post}^k$

From Structural Pricing to Regression Estimand

Zero-profit pricing (log-linearized): $r^k \approx (\rho + g) - m^k$, $m^k \equiv \ln \mu^k$

Four cells:

	Below	Above
Pre	$\rho + g_0 - m_{B,Pre}^k$	$\rho + g_0 + \Delta g^{HB} - m_{A,Pre}^k$
Post	$\rho + \Delta \rho + g_0 - m_{B,Post}^k$	$\rho + \Delta \rho + g_0 + \Delta g^{HB} + \Delta g^{reform} - m_{A,Post}^k$

Diff-in-Disc estimand $\hat{\beta}_3^k$:

$$\begin{aligned}
 \hat{\beta}_3^k &= (r_{A,Post}^k - r_{A,Pre}^k) - (r_{B,Post}^k - r_{B,Pre}^k) \\
 &= \Delta g^{reform} - \underbrace{\left[(m_{A,Post}^k - m_{B,Post}^k) - (m_{A,Pre}^k - m_{B,Pre}^k) \right]}_{= 0 \text{ under Local Parallel Trends}}
 \end{aligned}$$

Identifying assumption (Local Parallel Trends):

$$\hat{\beta}_3^k = \Delta g^{reform} \times \frac{1}{\mu^k}$$

Test: estimate $\hat{\beta}_3^k$ separately for $k \in \{\text{cosign, solo}\}$ and compare.

The Identification Problem: Endogenous Sorting

In a standard Regression Discontinuity (RD) design, identification requires continuity of potential outcomes at the cutoff.

The Violation:

- Borrowers have incentives to “bunch” below the limit to avoid fees.
- Sorting is **non-random**

Bias in Standard RD

Consequently, the standard Cross-Sectional RD estimator at time t yields:

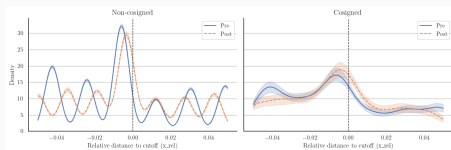
$$\tau_{SRD,t} = \tau_{\text{True}} + \underbrace{\left[\lim_{x \downarrow 0} E[Y(0)|x] - \lim_{x \uparrow 0} E[Y(0)|x] \right]}_{\text{Selection Bias } B_t \neq 0}$$

Key reference: **Difference-in-Discontinuities (Diff-in-Disc)** (Grembi, Nannicini, and Troiano, 2016).

Core Intuition: Relax the assumption that potential outcomes are continuous ($B_t = 0$) to assume the sorting is constant over time ($B_t = B_0$).

Validating Constant Sorting

Density: 2021 vs. 2022



Formal Test (McCrary)

Local linear model on binned log-densities:

$$y_{bt}^k = \alpha^k + \delta_1^k \text{Post} + \delta_2^k \mathbb{1}_{>0} + \beta^k (\mathbb{1}_{>0} \times \text{Post}_t)$$

$H_0: \beta^k = 0$ (jump constant over time)

Sample	$\hat{\beta}$ (SE)	p
Non-Cosigned	-4.57 (3.19)	0.17
Cosigned	-3.13 (2.85)	0.29

Fail to reject H_0 : sorting intensity unchanged post-reform for both groups.

Econometric Specification

To allow for heterogeneous effects and distinct sorting mechanisms, we estimate the model **separately** for each subsample $k \in \{\text{Cosigner}, \text{Non-Cosigner}\}$:

$$Y_{ilt}^k = \alpha^k + \beta_1^k D_{it} + \beta_2^k \text{Post}_t + \beta_3^k (D_{it} \times \text{Post}_t) + \lambda^k(\tilde{X}_{it}) + \mathbf{Z}_{it} \gamma^k + \delta_l^k + \epsilon_{ilt}^k$$

Definitions:

- $D_{it} = \mathbb{1}(\tilde{X}_{it} > 0)$: Above Conforming Limit indicator.
- β_{3k} : **Parameter of Interest** (Causal Effect for group k).
- $\lambda_k(\cdot)$: Flexible linear slopes (allowed to vary by group k).
- δ_l : Lender Fixed Effects (clustered SEs).

Hypothesis Testing: After estimation, we test for the statistical difference between the groups:

$$H_0 : \beta_3^{\text{Cosigner}} - \beta_3^{\text{Non-Cosigner}} = 0$$

Main Results: Rejection Rates

Table 5: Diff-in-Disc: Application Rejection Rate

	Rejection Decision		
	Non-Cosigner	Cosigner	Cos. – Non
	(1)	(2)	(3)
Above $\times$ Post	+1.48 (1.63)	–2.33 (6.73)	–3.82 (6.93)
Unconditional Mean: 7.57%			
County FE	✓	✓	
Lender FE	✓	✓	
Demographics	✓	✓	
Loan Controls	✓	✓	

Imbens-Kalyanaraman optimal bandwidth. SE clustered by lender. * $p < 0.10$,

** $p < 0.05$, *** $p < 0.01$.

Main Results: Interest Rates

Table 6: Diff-in-Disc: Mortgage Interest Rate (pp)

	Separate Regressions		
	Non-Cosigner	Cosigner	Cos. – Non
	(1)	(2)	(3)
Above $\times$ Post	–1.851*** (0.145)	–0.691*** (0.223)	+1.161*** (0.266)
Unconditional Mean: 4.11%			
County FE	✓	✓	
Lender FE	✓	✓	
Demographics	✓	✓	
Loan Controls	✓	✓	

Endogenous Sorting after Reform

Differential sorting on observables?, Endogenous subgroup selection?

Table 7: Borrower Composition Check

Outcome	Diff-in-Disc Estimate (β_3)		
	Non-Cosigner	Cosigner	Difference
Log Loan Amount	0.060*** (0.00)	0.048*** (0.00)	-0.012 (0.46)
Log Property Value	0.046*** (0.00)	0.040** (0.01)	-0.006 (0.72)
Log Income	-0.020 (0.39)	0.116 (0.23)	0.137 (0.17)
Unconditional Mean Log Loan Amount: 12.82			
Unconditional Mean Log property Value: 13.01			
Unconditional Mean Log Income: 11.56			
Change in Cosigning Probability (β_3^{Prob})			
-0.000*** (Unconditional Mean: 0.033)			

Estimates of β_3 using optimal bandwidths. Standard Errors are clustered at the lender level.

- **No Differential Sorting:** "Jumps" in Loan Amount, Property Value, and Income are statistically identical for both groups ($p > 0.10$).
- **No Sorting into Contracts:** Share of cosigned mortgage does not change.

Local Parallel Trends

Estimate the model at a “Placebo Cutoff” (80% of the actual limit) to test if the heterogeneous pricing arises naturally.

Outcome	Placebo Estimates ($\beta_3^{Placebo}$)		
	Non-Cosigner	Cosigner	Difference
Interest Rate (%)	0.563*** (0.00)	0.303* (0.06)	-0.260 (0.13)
Discount Points (\$)	266 (0.52)	1,035 (0.10)	769 (0.30)
App. Rejection	-0.023 (0.16)	-0.030 (0.24)	-0.007 (0.83)

Table 8: Estimates at a placebo cutoff (80% of CLL)

- **No statistically significant difference** between Cosigners and Non-Cosigners at the placebo threshold.
- For Interest Rates, the placebo effect is positive, whereas the effect at the real cutoff was negative.

Quantitative Model

Model Environment and Timing

Demographics and uncertainty

- Overlapping generations, lifecycle length $j = 1, \dots, J$.
- Idiosyncratic, uninsurable income shocks $z_j \in \mathcal{Z}$ follow a Markov process $\Pi(z'|z)$.

Assets and housing

- Risk-free asset $a \geq 0$ with return r .
- Rental housing size $h^R \in \mathcal{H}^R$ at price p^R .
- Owned housing size $h \in \mathcal{H}^O$ at price p^O .

Life events

- In final period, agent can bequeath assets.
- At age $j_B + 25$ (parental help age, $j_P = 43$): parents choose *no help* / *cosign* / *cash* / *both*.
- At age $J - 30$, agents receive an inheritance.
- At age $j_B = 18$ (buying age): household chooses rent vs. buy with mortgage.
- During mortgage term (M years): owners choose pay vs. default.

State Space (One Lifecycle Agent, Two Life Stages)

Same agent over the lifecycle: the “young/child” and “old/parent” problems are the same individual’s decisions at different ages. **State Space**

$$s \equiv (j, a, z, \theta, z^p, \omega, h, h^c, \eta)$$

- j = age, a = liquid assets, z = idiosyncratic productivity
- $\theta \in \{G, B\}$: private type (observed by agent, unobserved by bank);
 $\Pr(\theta = G \mid z) = \lambda_z(z)$
- z^p = parent’s productivity
- ω = housing status:

$$\omega \in \{RB, PD\}$$

- h = housing stock
- $h^c \in \{1, 2, 3, 4\}$: realized help received when young (1=none, 2=cosign, 3=cash, 4=both)
- $\eta \in \{1, 2, 3, 4\}$ = help commitment made when old (1=none, 2=cosign, 3=cash, 4=both)

Parental help received when young

Before it is realized, the young agent forms beliefs

$$\pi^c(z^p) = (\Pr(h^c = 2 \mid z^p), \Pr(h^c = 3 \mid z^p), \Pr(h^c = 4 \mid z^p)).$$

Recursive Objects (Same Agent, Stage-Dependent Problems)

Stage-dependent value functions of the same agent

$$V_{RB}(s), \quad V_{PD}(s)$$

- $V_{RB}(s)$: renter/buyer stage (young ages, tenure choice)
- $V_{PD}(s)$: homeowner stage (mortgage repayment vs default)

Help decision when old (same agent later in life)

At age $j = j_P = 43$, the agent becomes a parent and chooses

$$\eta \in \{1, 2, 3, 4\} \quad (\text{none/cosign/cash/both})$$

which determines the help regime faced by their own child in the next generation:

$$\eta \longrightarrow h^c \text{ for the child.}$$

Interpretation

- The only difference between “young” and “old” is **the age index j** (and hence which constraints/choices are active).
- Intergenerational linkage enters through z^P and η .

Young Stage: Rent vs. Buy

At the buying age $j = j_B = 18$, with i.i.d. Type-I extreme value taste shocks $\varepsilon^R, \varepsilon^B$ scaled by σ_{RB} :

$$V_{RB}(s) = \mathbb{E}_\varepsilon \left[\max \left\{ V_{RB}^{rent}(s) + \sigma_{RB} \varepsilon^R, V_{RB}^{buy}(s) + \sigma_{RB} \varepsilon^B \right\} \right], \quad s = (j, a, z, z^P, \omega, h^c, \eta).$$

Equivalently (log-sum):

$$V_{RB}(s) = \sigma_{RB} \log \left(\exp \left(\frac{V_{RB}^{rent}(s)}{\sigma_{RB}} \right) + \exp \left(\frac{V_{RB}^{buy}(s)}{\sigma_{RB}} \right) \right).$$

Buying

$$V_{RB}^{buy}(s) = \max_{a', h \in \mathcal{H}^O} \left\{ u(c, h, 1) + \beta \mathbb{E} [V_{PD}(s')] \right\}$$

subject to

$$c = we(j, z) + (1 + r)a - a' + \underbrace{(\lambda_m p^O h - p^O h)}_{\text{downpayment} = -(1 - \lambda_m) p^O h} + \underbrace{T(z^P)}_{\text{parental cash transfer if } h^c = 3}.$$

Renter

Default

Mortgage Access and Payment Construction

Mortgage contract (purchase of house size h at price p^O)

$$\text{House value} = p^O h, \quad L = p^O h - (1 - \lambda_m) p^O h$$

- L = loan principal
- $DP(h^c)$ depends on help regime: none / cosign / cash

Fixed-rate annuity payment (maturity M)

$$m(L, R(z), M) = \frac{R(z)(1 + R(z))^M}{(1 + R(z))^M - 1} \cdot L.$$

Mortgage access (reduced form PTI screening)

$$\text{PTI}(h, h^c, z, z^p) \equiv \frac{m(L, M)}{w e(j_B, z) + \mathbf{1}\{h^c = 2\} w e(j_P, z^p)}.$$

$$\text{Approved} \iff \text{PTI}(h, h^c, z, z^p) \leq \overline{\text{PTI}}.$$

Key link: cosigning affects only access decision.

Old Stage: Irreversible Help Commitment (η)

At age $j = j_P$, the parent makes a **one-time, irreversible** help commitment.

Parent problem (commitment choice with logit shocks)

$$V_P(j_P, a, z, \omega, h, \eta = 1) = \max_{\eta \in \{1, 2, 3\}} \left\{ \tilde{V}_P(j_P, a, z, \omega, h; \eta) + \varepsilon^\eta \right\},$$

where $\varepsilon^{\eta'}$ are i.i.d. Type-I Extreme Value (Gumbel) shocks with scale $\sigma_{\text{help}} > 0$.

How commitment enters the parent budget (flow costs)

$$c = we(j, z) + (1 + r)a - a' - \text{housing} - \underbrace{\mathbf{1}\{\eta = 2\} \ell(\cdot)}_{\text{cosigning}} - \underbrace{\mathbf{1}\{\eta = 3\} \kappa(\cdot)}_{\text{cash}}.$$

Cosigning as expected liability (insurance-like stream)

The cosigning burden depends on the child's mortgage risk and size:

$\ell(z^c, h_c)$ increases in leverage/house size h_c and decreases in z^c .

Child's portfolio is unobserved at commitment, so the parent internalizes the **expected liability**

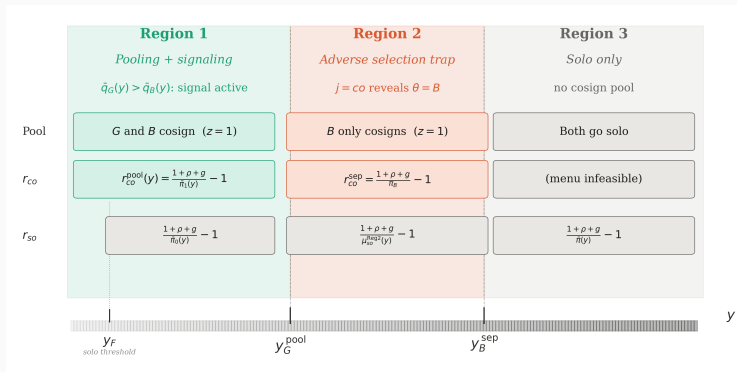
$$\ell(\cdot) \equiv \mathbb{E}[\ell(z^c, h_c) \mid z^P, z^c].$$

Table 9: Baseline Calibration

Parameter	Symbol	Value	Source
<i>Preferences</i>			
Discount factor	β	0.96	Various
Risk aversion	σ	1.5	Various
Housing share	θ	0.25	Garriga & Hedlund (2022)
Ownership taste premium	ν	1.1	Garriga & Hedlund (2022)
Bequest strength	ϕ_1	9.5	De Nardi (2004)
Bequest threshold	ϕ_2	11.6	De Nardi (2004)
<i>Income process</i>			
Persistence	ρ	0.96	Storesletten, Telmer & Yaron (2004)
Innovation s.d.	σ_ϵ	0.12	Various
<i>Housing & prices</i>			
Rental price	p^R	$0.09 \cdot p^O$	CBRE
<i>Mortgage contract & borrowing</i>			
Baseline LTV (10% down)	λ_m	0.90	Fannie Mae
Maturity	M	30	Fannie Mae
Interest rate (risk-free)	r	0.03	Various

The Three Region Visualized

Figure 6: Model Solution: Three Regions of the State Space



Targeted & Untargeted Moments

Moment		Target	Model	Source
<i>Targeted (in SMM loss)</i>				
M1	Homeownership rate (ages 19–49)	39.4%	30.6%	PSID/CPS
M2	Mortgage default rate	3.74%	5.23%	NY Fed
M3	Share of purchases with parental cosigning	5.47%	3.82%	PSID
M4	Share of purchases with parental cash gift	35.0%	32.3%	BofA/YouGov
M5	Cosigning income gradient: $(\bar{s}_{Q4} - \bar{s}_{Q1})/\bar{s}$	−0.148	−1.049	PSID
M6	Q1 default rate deviation: $(d_{Q1} - \bar{d})/\bar{d}$	0.667	2.762	NY Fed
M7	Cosign rate premium: $\bar{r}_{co} - \bar{r}_{non}$	0.064	0.017	HMDA
M8	Rate pass-through DiD: $\hat{\beta}_3^{co} - \hat{\beta}_3^{non}$	0.77 pp	0.179 pp	HMDA
<i>External Validation</i>				
M9	Rejection DiD: $\hat{\beta}_3^{co} - \hat{\beta}_3^{non}$	−3.8 pp	2.8 pp	HMDA

Counterfactual: Joint-Income Risk Pricing

Policy question: what if lenders used *joint* parent–child income to determine the risk bucket for cosigned mortgages, instead of child income only?

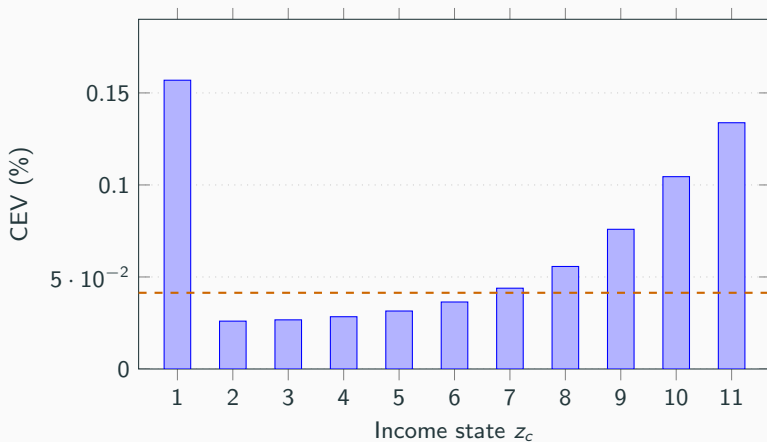
- Decoupling reversed - reduces adverse selection

Results:

- Cosigning rates increase by 40% relative to baseline
- This closes 22% of the cosigning gap to the average international comparison
- Homeownership rates increase by about 1% - concentrated in around median
- Mean Wealth increases by 1.2%
- No increased default rates.

Behind the scenes:

Welfare: Consumption Equivalent Variation by Income State



Conclusion

1. Parental cosigning lower in US than comparable economies.
2. More than 20% of difference can be explained by institutional decoupling.
3. Decoupling creates adverse selection and reduces homeownership - parental signal not function everywhere in the economy.
4. Reform that re-couples risk assessment reduces adverse selection and increases homeownership without increasing default.

Appendix

Robustness of Correlations

Table 10: Parental Cosigning: Extensive and Intensive Margins

	Fact 1 Rejection Rate
Parental cosigning	-0.0154*** (0.0017)
Lender FE	Yes
State FE	Yes
Year FE	Yes
Demographics	Yes
Property Value & Loan Amount	Yes
R^2	0.009
Observations	6,672,649

Rejection Rates

Interest Rates

Prop. Value

Loan Amount

◀ Back to Fact 1

Robustness of Correlations

Table 10: Parental Cosigning: Extensive and Intensive Margins

	Fact 1 Rejection Rate	Fact 2 Log Property Value
Parental cosigning	-0.0154*** (0.0017)	0.0818*** (0.0027)
Lender FE	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Demographics	Yes	Yes
Property Value & Loan Amount	Yes	—
R^2	0.009	0.180
Observations	6,672,649	5,795,691

Rejection Rates

Interest Rates

Prop. Value

Loan Amount

◀ Back to Fact 1

Robustness of Correlations

Table 10: Parental Cosigning: Extensive and Intensive Margins

	Fact 1 Rejection Rate	Fact 2 Log Property Value	Fact 2 Log Loan Amount
Parental cosigning	-0.0154*** (0.0017)	0.0818*** (0.0027)	0.0510*** (0.0030)
Lender FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Property Value & Loan Amount	Yes	—	—
R^2	0.009	0.180	0.163
Observations	6,672,649	5,795,691	5,795,691

Rejection Rates

Interest Rates

Prop. Value

Loan Amount

◀ Back to Fact 1

Robustness of Correlations

Table 10: Parental Cosigning: Extensive and Intensive Margins

	Fact 1 Rejection Rate	Fact 2 Log Property Value	Fact 2 Log Loan Amount	Fact 4 Interest Rate
Parental cosigning	-0.0154*** (0.0017)	0.0818*** (0.0027)	0.0510*** (0.0030)	-0.0161 (0.0103)
Lender FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Property Value & Loan Amount	Yes	—	—	Yes
R^2	0.009	0.180	0.163	0.000
Observations	6,672,649	5,795,691	5,795,691	5,790,238

Rejection Rates

Interest Rates

Prop. Value

Loan Amount

◀ Back to Fact 1

Table 11: Cosigning and Intensive Margin (Webscraping)

Log Property Value	
Parental cosigning	0.1200*** (0.0063)
County FE	Yes
Lender FE	No
State FE	No
Year FE	Yes
Demographics	Yes
R^2	0.562
Observations	158,354

Table 11: Cosigning and Intensive Margin (Webscraping)

	Log Property Value	Log Loan Amount
Parental cosigning	0.1200*** (0.0063)	0.0988*** (0.0066)
County FE	Yes	Yes
Lender FE	No	No
State FE	No	No
Year FE	Yes	Yes
Demographics	Yes	Yes
R^2	0.562	0.481
Observations	158,354	158,354

Table 11: Cosigning and Intensive Margin (Webscraping)

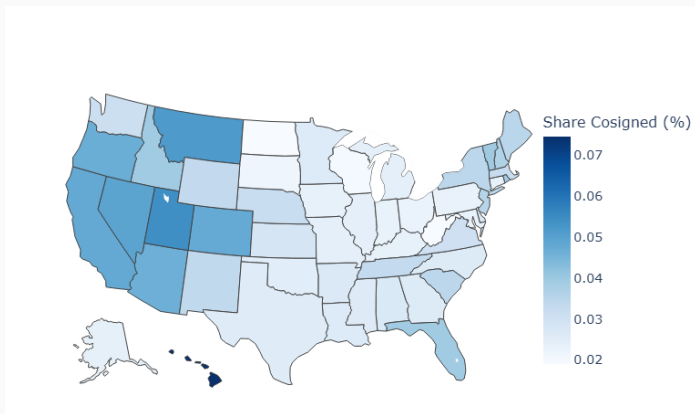
	Log Property Value	Log Loan Amount	Interest Rate
Parental cosigning	0.1200*** (0.0063)	0.0988*** (0.0066)	0.1045*** (0.0175)
County FE	Yes	Yes	Yes
Lender FE	No	No	No
State FE	No	No	No
Year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
R^2	0.562	0.481	0.398
Observations	158,354	158,354	143,307

Table 12: Diff-in-Disc: Discount Points Paid (\$)

	Separate Regressions		
	Non-Cosigner	Cosigner	Cos.—Non
	(1)	(2)	(3)
Above $\times$ Post	−8,756** (3,630)	+216 (1,328)	+8,972** (3,866)
County FE	✓	✓	
Lender FE	✓	✓	
Demographics	✓	✓	

Variety across states

Figure 7: Share of Cosigned Mortgages, 2024



Property Prices - Larger Counties

Figure 8: Property Prices - At least 100 Mortgages per Year

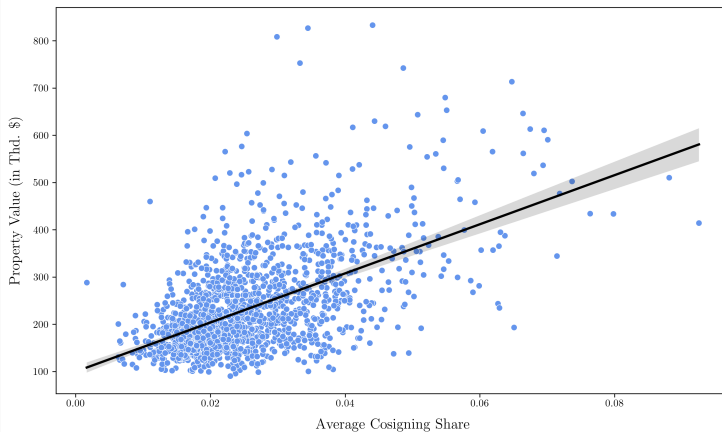


Table 13: Extensive Margin & Cosigning - Adding Control

	Application Rejected		
Parental cosign	-0.0150*** (0.0017)	-0.0434*** (0.0019)	-0.0154*** (0.0017)
County FE	No	Yes	No
Lender FE	Yes	No	Yes
State FE	No	No	No
Year FE	No	No	Yes
Demographics	Yes	Yes	Yes
Property Value & Loan Amount	Yes	Yes	Yes
R^2	0.009	0.086	0.009

Table 14: Parental Cosigning on Mortgage Interest Rate

	Mortgage Interest Rate			
Parental cosigning	-0.0576*** (0.014)	-0.0964*** (0.011)	-0.0224** (0.0110)	-0.0415*** (0.0103)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.001	0.012	0.000	0.000
Observations	5 437 803	5 437 803	5 437 803	5 437 803

Table 15: Parental Cosigning on Loan Amount

Log Loan Amount				
Parental cosigning	0.0591*** (0.0027)	0.0632*** (0.0027)	0.0579*** (0.0030)	0.0380*** (0.0030)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.297	0.151	0.280	0.084
Observations	5,705,601	5,705,601	5,705,601	5,705,601

Table 16: Parental Cosigning and Property Value

Log Property Value				
Parental cosigning	0.0900*** (0.0027)	0.0985*** (0.0027)	0.0889*** (0.0027)	0.0681*** (0.0025)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.324	0.163	0.304	0.093
Observations	5,705,601	5,705,601	5,705,601	5,705,601

Who are the children?

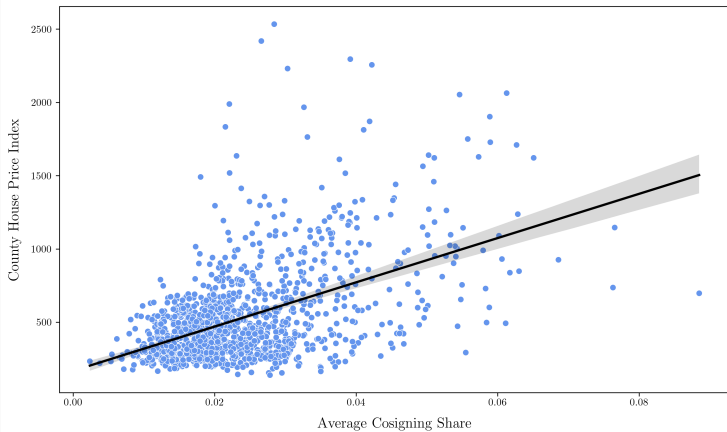
Table 17: Who are the children? - Some Means

	Savings	Total Wealth	Income**	Years Educ***
Not Cosigners	40,024	379,534	166,779	15.23
Cosigners	36,644	408,635	150,694	14.42

[Back](#)

Property Price Index

Figure 9: House Price Index



Guarantee Fees I

Figure 10: Guarantee Fees and Delinquency rates

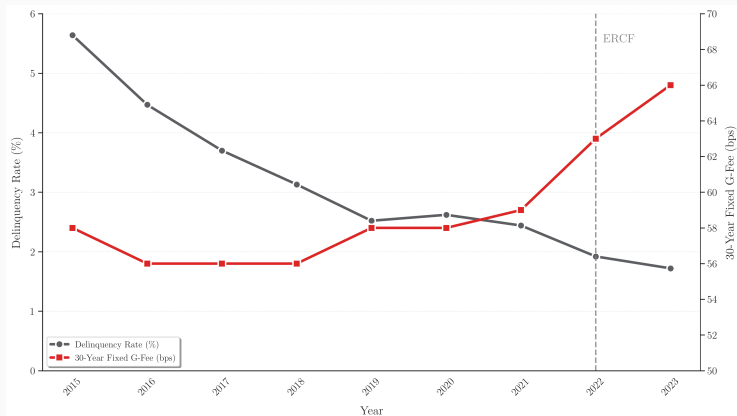
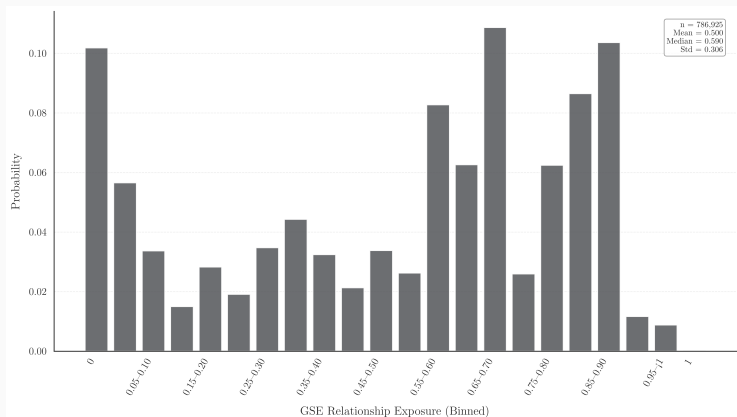


Figure 11: GSE Exposure



Guarantee Fees II

Figure 12: Guarantee Rates and Interest Rates

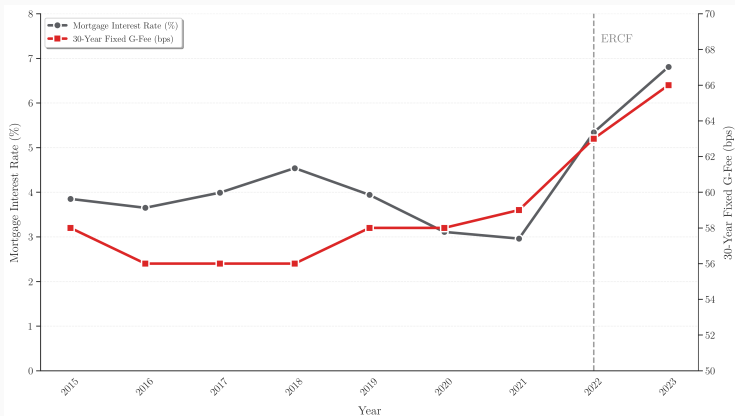
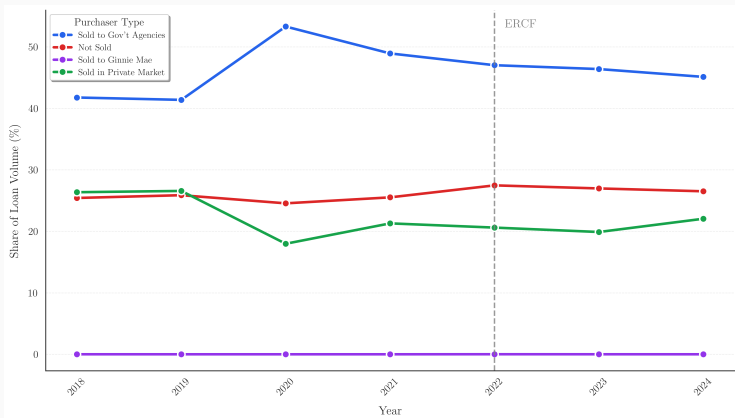


Figure 13: Loan Shares



Crucial Assumption: Time-Invariant Sorting Bias

Assumption 1 (Local Parallel Trends)

The discontinuity in potential outcomes absent the treatment is invariant across time.

$$\underbrace{B_1}_{\text{Post-Bias}} = \underbrace{B_0}_{\text{Pre-Bias}}$$

Formal Proposition:

$$\begin{aligned} & \left(\lim_{x \downarrow 0} E[Y(0)|x, t = 1] - \lim_{x \uparrow 0} E[Y(0)|x, t = 1] \right) \\ &= \left(\lim_{x \downarrow 0} E[Y(0)|x, t = 0] - \lim_{x \uparrow 0} E[Y(0)|x, t = 0] \right) \end{aligned}$$

Implication: If the composition of borrowers at the threshold does not change differentially over time, B_0 is a valid counterfactual for B_1 .

Robustness: Sensitivity to Bandwidth (50k–70k)

Table 18: Bandwidth Robustness

Outcome (β_3)	(1) h = 50,000	(2) h = 60,000	(3) h = 70,000
Panel A: Interest Rate (%)			
Non-Cosigner	-1.051***	-0.889***	-0.788***
Cosigner	-0.566***	-0.311*	-0.374**
<i>Difference</i>	0.486**	0.579***	0.414**
Panel B: Discount Points (\$)			
Non-Cosigner	-319	-135	-52
Cosigner	878	1,527*	997
<i>Difference</i>	1,197	1,662*	1,049
Panel C: Application Rejection			
Non-Cosigner	0.012	0.014*	0.010
Cosigner	-0.044	-0.060	-0.044
<i>Difference</i>	-0.056	-0.074	-0.054

Comparison of Diff-in-Disc estimates across fixed bandwidths.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Renting

$$V_{RB}^{rent}(s) = \max_{a', h^R \in \mathcal{H}^R} \left\{ u(c, h^R, 0) + \beta \mathbb{E} [V_{RB}(s')] \right\}$$

subject to

$$c = we(j, z) + (1 + r)a - a' - p^R h^R.$$

Back

Homeowner Stage: Pay vs. Default (with Extreme Value Shocks)

For $j \in \{j_B + 1, \dots, j_B + M\}$, owners solve

$$V_{PD}(s) = \max\{V_{PD}^{pay}(s), V_{PD}^{def}(s)\}.$$

Pay

$$V_{PD}^{pay}(s) = \max_{a'} \{u(c, h, 1) + \beta \mathbb{E}[V_{PD}(s')]\} + \varepsilon^{pay}$$

subject to

$$c = we(j, z) + (1 + r)a - a' - m(h, z).$$

Default

$$V_{PD}^{def}(s) = \max_{a'} \{u(c, h, 1) - \xi + \beta \mathbb{E}[V_{\text{next}}(s')]\} + \varepsilon^{def}$$

subject to

$$c = we(j, z) + (1 + r)a - a'.$$

Extreme value shocks (logit)

$\varepsilon^{pay}, \varepsilon^{def}$ i.i.d. Type-I Extreme Value (Gumbel).

Then the conditional choice probabilities are

$$\exp\left(\frac{\tilde{V}_{PD}^{pay}(s)}{\sigma_{pay}}\right)$$

Per-period utility (consumption and housing services)

$$u(c, h, \text{owner}) = \frac{(c^{1-\theta} \cdot (\chi(\text{owner})h)^\theta)^{1-\sigma} - 1}{1-\sigma}.$$

- $\chi(\text{owner}) > 1$ captures an ownership premium in housing services.
- Discount factor β ; CRRA curvature σ ; housing share θ .

Terminal Period: Bequests and Intergenerational Wealth

At age $j = J$ the agent consumes and leaves bequests:

$$V(J, s) = u(c, h, \text{owner}) + b(a', h).$$

Terminal budget

$$c = we(J, z) + (1 + r)a - a' - \text{housing outlays}.$$

Bequest motive

$$b(a', h) = \phi_1 \cdot \frac{(a' + p^O h + \phi_2)^{1-\sigma} - 1}{1 - \sigma}.$$

Link to the paper: cosigning affects early-life housing access and leverage, shaping lifecycle portfolios and wealth transmission.

Market Clearing and Equilibrium I

Prices are taken as given: (r, w, p^O, p^R) are exogenous.

State space

$$s \equiv (j, a, z, z^P, \omega, h, h^c, \eta),$$

where j is age, a liquid assets, z productivity, z^P parental type, $\omega \in \{RB, PD\}$ housing stage, h owned housing (if any), $h^c \in \{1, 2, 3, 4\}$ help received (1=none, 2=cosign, 3=cash, 4=both), and $\eta \in \{1, 2, 3, 4\}$ irreversible help commitment made when old.

Optimal policy functions

Given prices and mortgage rules, the solution delivers:

$$a' = g_a(s), \quad h' = g_h(s), \quad \omega' = g_\omega(s), \quad d = g_d(s), \quad \eta' = g_\eta(s),$$

where $d \in \{0, 1\}$ is the default decision and $\eta' = \eta$ after commitment (irreversible).

1) Stationary distribution (consistency)

Let μ denote the cross-sectional distribution over s .

$$\mu' = \mathcal{T}(\mu; \{g_a, g_h, g_\omega, g_d, g_\eta\}, \Pi), \quad \mu' = \mu.$$

Market Clearing and Equilibrium II

2) Default consistency $\forall (z^c, z^p, h)$

$$\Pr^{\text{guess}}(\text{default} \mid z^c, z^p, h) = \Pr^{\text{implied}}(\text{default} \mid z^c, z^p, h; \mu).$$

3) Cash-transfer consistency $\forall (z^p, z^c, h)$

$$\pi_{\text{cash},h}^{\text{guess}}(z^p, z^c, h) = \pi_{\text{cash},h}^{\text{implied}}(z^p, z^c, h; \mu).$$

4) Cosigning consistency $\forall (z^p, z^c, h)$

$$\pi_{\text{cosign},h}^{\text{guess}}(z^p, z^c, h) = \pi_{\text{cosign},h}^{\text{implied}}(z^p, z^c, h; \mu).$$

5) Inheritance consistency $\forall (z^c, z^p)$

$$\text{inh}^{\text{guess}}(z^c, z^p) = \text{inh}^{\text{implied}}(z^c, z^p; \mu), \quad \text{inh}^{\text{implied}}(z^c, z^p; \mu) = \bar{b}(z^p; \mu) \cdot \text{scale}(z^c, z^p).$$

Equilibrium: policy functions $\{g_a, g_h, g_\omega, g_d, g_\eta\}$ and a stationary distribution μ such that the implied conditional moments under μ coincide with the belief objects

$\{\Pr(\text{default}), \pi_{\text{cash},h}, \pi_{\text{cosign},h}, \text{inh}\}$. [Back](#)

Table 19: Baseline Calibration

Parameter	Symbol	Value	Source
<i>Discrete-choice (EV shocks)</i>			
Rent–buy logit scale	$\sigma_{\text{rent/buy}}$	0.3	
Help-type logit scale	σ_{help}	0.1	
Default logit scale	σ_{default}	0.1	
Default stigma	$-\xi$	7.0	

Main Results: Rejection Rates (w/ Controls)

Table 20: Diff-in-Disc: Application Rejection Rate (w/ additional controls)

	Rejection Decision		
	Non-Cosigner	Cosigner	Cos.—Non
	(1)	(2)	(3)
Above $\times$ Post	+1.17 (1.40)	−4.36 (4.41)	−5.52 (4.61)

Unconditional Mean: 7.57%

State FE	✓	✓
Lender FE	✓	✓
Demographics	✓	✓
Loan Controls	✓	✓

Main Results: Interest Rates (w/ Controls)

Table 21: Diff-in-Disc: Mortgage Interest Rate (pp, w/ additional controls)

	Separate Regressions		
	Non-Cosigner	Cosigner	Cos.—Non
	(1)	(2)	(3)
Above $\times$ Post	−1.382*** (0.142)	−0.606*** (0.230)	+0.776*** (0.270)
Unconditional Mean: 4.11%			
State FE	✓	✓	
Lender FE	✓	✓	
Demographics	✓	✓	
Loan Controls	✓	✓	

Proof: $z=1$ is a Credible Signal

Goal: Show $\bar{q}_G(y) > \bar{q}_B(y)$, so Bayes' rule gives $\Lambda_1(y) > \lambda(y)$.

Parent's net surplus from offering the guarantee:

$$\Delta U_p(\theta; y, y_p) \equiv \beta \Delta V(y, \theta) - (1 - \pi_\theta) \Delta u_p.$$

Taking the difference between Good- and Bad-type parents (using $\Delta V(y, \theta) = \pi_\theta(u_{co} - u_{so}) + (1 - \pi_\theta)(\underline{u}_{co} - \underline{u}_{so})$):

$$\Delta U_p(G) - \Delta U_p(B) = \Delta\pi(\Delta u_p - \beta[(u_{so} - u_{co}) + (\underline{u}_{co} - \underline{u}_{so})]).$$

Since $\Delta\pi > 0$, **Cost Dominance** (Assumption 1) ensures the parenthesized term is strictly positive for all (y, y_p) in the support:

$$\Delta U_p(G; y, y_p) > \Delta U_p(B; y, y_p) \quad \forall (y, y_p).$$

From surplus to cosigning rates: F_η strictly increasing $\Rightarrow$ pointwise $q_G(y, y_p) > q_B(y, y_p)$. Integrating over $F(y_p | y)$:

$$\bar{q}_G(y) = \int q_G(y, y_p) dF(y_p | y) > \int q_B(y, y_p) dF(y_p | y) = \bar{q}_B(y).$$

Bayesian update: $\bar{q}_G > \bar{q}_B$ and Bayes' rule give $\Lambda_1(y) > \lambda(y) > \Lambda_0(y)$: the guarantee is a credible positive signal of creditworthiness. $\square$

Proof: Three-Region Equilibrium I — Rates, Sorting, Cutoffs

Step 1: Unique break-even rates.

- *Region 2*: Bank knows $\theta = B$ (full revelation), so $r_{co}^{sep} = (1 + \rho + g)/\pi_B - 1$ directly.
- *Region 1*: Bank's expected revenue $ER(r_{co}) = \mu_{co}(r_{co}, y)(1 + r_{co})$ is strictly increasing in r_{co} under (A3b) ($\bar{f}$ small), since $\partial\mu_{co}/\partial r_{co} < 0$ but bounded by $K < \mu_{co}/(1 + r_{co})$. By IVT + strict monotonicity, unique $r_{co}^{pool}(y)$ exists and is C^1 in y (Implicit Function Theorem).

Step 2: ΔV strictly decreasing in y . $d\Delta V/dy = \pi_\theta dG/dy < 0$ for $y > y^{**}$ (Wealth Effect Lemma). The solo rate r_{so} drops *discontinuously* at y_G^{pool} (Good types flood the solo pool), causing G to jump *downward*, reinforcing exit. No upward crossings exist.

Step 3: Existence and uniqueness of cutoffs. By (A2): $\Delta V(y, G) > 0$ near y_F^+ and $\rightarrow -\infty$. Define:

$$y_G^{pool} \equiv \inf\{y : \Delta V(y, G; r_{co}^{pool}(y)) \leq 0\}, \quad y_B^{sep} \equiv \inf\{y > y_G^{pool} : \Delta V(y, B; r_{co}^{sep}) \leq 0\}.$$

Strict monotonicity (Step 2) + infimum construction handle both the continuous case ($\Delta V = 0$) and the discontinuous case (ΔV jumps below zero). Uniqueness of each cutoff follows from the absence of upward crossings.

Proof: Three-Region Equilibrium II — Consistency, Optimality, Uniqueness

Step 4: Belief consistency (Bayes' rule). On the equilibrium path, Bayes' rule recovers equilibrium beliefs exactly in each region: $\mu_{co} = \bar{\pi}_1(y)$ (R1), $\mu_{co} = \pi_B$ (R2), $\mu_{so} = \bar{\pi}_0(y)$ (R1), $\mu_{so} = \mu_{so}^{Reg2}(y) > \bar{\pi}(y)$ (R2), $\mu_{so} = \bar{\pi}(y)$ (R3). ✓

Step 5: Borrower optimality.

- *Region 1:* $\Delta V(y, \theta; r_{co}^{pool}) > 0$ for both types $\Rightarrow$ all $z = 1$ borrowers choose cosign.
- *Region 2:* Good types: $\Delta V(y, G; r_{co}^{sep}) < \Delta V(y, G; r_{co}^{pool}) \leq 0 \Rightarrow$ prefer solo.
Bad types: $\Delta V(y, B; r_{co}^{sep}) > 0 \Rightarrow$ prefer cosign.
- *Region 3:* $\Delta V(y, B; r_{co}^{sep}) \leq 0$ and SC lemma gives $\Delta V(y, G) < \Delta V(y, B) \leq 0 \Rightarrow$ both prefer solo. □

Step 6: Off-path deterrence. Off-path belief $\mu_{co}^{op} = \pi_B$ (worst case for deviator) deters all deviations: cosign pays weakly less than solo at every income above y_G^{pool} . Parental offers of $z = 1$ in Region 3 impose liability $(1 - \pi_\theta)\Delta u_p > 0$ with no benefit. ✓

Step 7: Uniqueness. Rate schedule uniquely pinned by zero-profit + beliefs (Step 1). Cutoffs uniquely pinned by strict monotonicity of ΔV (Steps 2–3). On-path beliefs uniquely pinned by Bayes' rule (Step 4). Off-path belief $\mu_{co}^{op} = \pi_B$ is the only belief consistent with equilibrium sorting. No alternative belief supports a different cutoff pair. □

Proof: Quality Dominance Decomposition ($\bar{\mu}_{so} > \bar{\mu}_{co}$)

Goal: Show $\bar{\mu}_{so} - \bar{\mu}_{co} > 0$ by decomposing the pool-average quality gap.

Adding and subtracting the unconditional benchmark $\bar{\pi}(y)$ within each pool integral and substituting equilibrium beliefs in each region:

$$\begin{aligned} \bar{\mu}_{so} - \bar{\mu}_{co} = & \underbrace{\omega^{hi} \int_{y_G^{pool}}^{\infty} \bar{\pi}(y) f_{so}^{hi}(y) dy}_{B > 0} + \underbrace{\omega^{lo} \int_0^{y_G^{pool}} [\bar{\pi}_0(y) - \bar{\pi}(y)] f_{so}^{lo}(y) dy}_{D_2 > 0} + \underbrace{\omega^{lo} \int_0^{y_G^{pool}} \bar{\pi}(y) f_{so}^{lo}(y) dy}_{\text{common}} \\ & + \underbrace{\int_0^{y_G^{pool}} [\bar{\pi}(y) - \bar{\pi}_1(y)] f_{co}^1(y) dy}_{D_1 < 0} + \underbrace{\int_{y_G^{pool}}^{y_B^{sep}} [\bar{\pi}(y) - \pi_B] f_{co}^2(y) dy}_{A > 0}. \end{aligned}$$

Economic interpretation of each term:

- B : high-income solo borrowers (Region 3) carry unconditional quality $\bar{\pi}(y)$ — unambiguously positive.
- D_2 : in Region 1, solo choosers are $z = 0$ types, so $\bar{\pi}_0(y) > \bar{\pi}(y)$ — positive selection into solo.
- D_1 : in Region 1, cosign pool is enriched in Good types, so $\bar{\pi}_1(y) > \bar{\pi}(y)$ — this *reduces* the gap.
- A : in Region 2, the cosign pool is pure Bad types ($\pi_B < \bar{\pi}(y)$) — the adverse selection trap, strictly positive contribution.

Proof: Pass-Through Derivation and Quality Channel

Rejection pass-through formula. Let $p_j(g) = 1 - F_{DTI}^j(\kappa; g)$. A rise $dg > 0$ shifts each borrower's DTI by $\sigma_j dg$, pushing mass above κ . By the co-area formula:

$$\beta_{rej}^j \equiv \frac{\partial p_j}{\partial g} = \underbrace{\mathbb{E}^j[\sigma_j \mid DTI_j = \kappa]}_{\text{avg. rate sensitivity at margin}} \times \underbrace{f_{DTI}^j(\kappa; g)}_{\text{density at margin}}.$$

Quality and insulation channels. The two factors work in opposite directions for cosigners:

- *Quality channel:* $\sigma_j = L/(Y_j \mu_j(y))$ is higher when pool quality μ_j is lower. Adverse selection $\Rightarrow \bar{\mu}_{co} < \bar{\mu}_{so}$, so $\mathbb{E}[\sigma_{co}] > \mathbb{E}[\sigma_{so}]$ (under zero covariance of L/Y_j and $1/\mu_j$). This **amplifies** rate pass-through for cosigners.
- *Insulation channel:* $Y_{co} = y + y_p \gg y = Y_{so}$ shifts the cosign DTI distribution far below κ , so $f_{DTI}^{co}(\kappa) \ll f_{DTI}^{so}(\kappa)$. This **insulates** cosigners from rejection pass-through.

Rate pass-through: $\beta_{rate}^j = \mathbb{E}[1/\mu_j \mid j]$. Since only the quality channel applies (no DTI density factor), $\bar{\mu}_{co} < \bar{\mu}_{so}$ directly gives $\beta_{rate}^{co} > \beta_{rate}^{so}$.

Proof: Log-Normal Verification of Threshold Sparsity

Setup. Assume $DTI_j(g) \sim \mathcal{LN}(\nu_j, \tau^2)$ within each pool. The density ratio at rejection threshold κ is (exact, by difference-of-squares identity):

$$\frac{f_{DTI}^{co}(\kappa)}{f_{DTI}^{so}(\kappa)} = \exp\left(-\frac{(\nu_{so} - \nu_{co})(\ln \kappa - \bar{\nu})}{\tau^2}\right), \quad \bar{\nu} = \frac{\nu_{so} + \nu_{co}}{2}.$$

Condition (C2) — Threshold Sparsity. Substituting into the sparsity condition and using the zero-covariance and Jensen approximations ($\mathbb{E}[1/\mu_j] \geq 1/\bar{\mu}_j$, so replacing by $1/\bar{\mu}_j$ tightens the condition):

$$\exp\left(\frac{(\nu_{so} - \nu_{co})(\ln \kappa - \bar{\nu})}{\tau^2}\right) > \frac{\bar{\mu}_{so}}{\bar{\mu}_{co}} \cdot \frac{\mathbb{E}[L/Y_{so} \mid so]}{\mathbb{E}[L/Y_{co} \mid co]}.$$

The left-hand side grows with the mean log-DTI gap $\nu_{so} - \nu_{co}$ (assuming $\ln \kappa > \bar{\nu}$). The condition holds when the parental income buffer y_p is large enough to shift ν_{co} sufficiently below ν_{so} : the cosign DTI distribution sits far from the rejection threshold, confirming threshold sparsity in the empirical range. $\square$

Proof: Rejection Pass-Through Formula

Goal: Derive $\beta_{rej}^j = \mathbb{E}^j[\sigma_j \mid DTI_j = \kappa] \cdot f_{DTI}^j(\kappa; g)$.

Write $p_j(g) = 1 - F_{DTI}^j(\kappa; g)$ where $F_{DTI}^j(\kappa; g) = \Pr(DTI_j(g) \leq \kappa)$. A rise $dg > 0$ shifts each borrower's DTI by $\sigma_j dg$ (via the pricing equation), pushing mass rightward past κ , so p_j rises.

Formal derivation. Differentiate the expectation of the indicator with respect to g . Because $DTI_j(g)$ depends smoothly on g , differentiating the indicator function brings in a Dirac delta at the threshold:

$$\frac{\partial F_{DTI}^j}{\partial g} = -\mathbb{E}^j[\delta(DTI_j(g) - \kappa) \cdot \sigma_j] .$$

Applying the co-area formula ($\mathbb{E}[h(X) \delta(X - c)] = h(c) f_X(c)$) with $X = DTI_j(g)$, $c = \kappa$, $h = \sigma_j$:

$$\frac{\partial F_{DTI}^j}{\partial g} = -\mathbb{E}^j[\sigma_j \mid DTI_j(g) = \kappa] \cdot f_{DTI}^j(\kappa; g) .$$

Since $\beta_{rej}^j = \partial p_j / \partial g = -\partial F_{DTI}^j / \partial g$:

$$\beta_{rej}^j = \underbrace{\mathbb{E}^j[\sigma_j \mid DTI_j(g) = \kappa]}_{\text{avg. rate sensitivity at rejection margin}} \times \underbrace{f_{DTI}^j(\kappa; g)}_{\text{density of borrowers at margin}} . \quad \square$$

Insulation: $Y_{co} = y + y_p \gg Y_{so} = y$ shifts the cosign DTI distribution far below κ , so

Proof: Wealth Effect Lemma ($dG/dy < 0$ for $y > y^{**}$)

Goal: As income rises above y^{**} , the repayment-state gain $G(y) = u_{so}(y) - u_{co}(y)$ falls.

Direct income effect ($\partial G/\partial y = u_c^* - u_c$):

- *Step 1 (Housing)*. Solo loan $L_{so}(y) \propto y$ grows freely; cosign loan capped at L^* . Hence $H_{so}(y) > \alpha L^* = H^*$ for all $y > y^{**}$.
- *Step 2 (Consumption)*. At binding DTI: $c_{co}(y) - c_{so}(y) = \kappa y - (1 + r_{co})L^* > 0$. Cosign borrower has strictly more consumption.
- *Step 3 (Marginal utility)*. Decompose:

$$u_c(c_{so}, H_{so}) - u_c^*(c_{co}, H^*) = \underbrace{[u_c(c_{so}, H_{so}) - u_c(c_{co}, H_{so})]}_{\geq 0 \text{ (concavity, Step 2)}} + \underbrace{[u_c(c_{co}, H_{so}) - u_c(c_{co}, H^*)]}_{\geq 0 \text{ (} u_{ch} \geq 0, \text{ Step 1)}} >$$

Hence $\partial G/\partial y = u_c^* - u_c < 0$.

Indirect rate effects ($dr_{co}/dy, dr_{so}/dy$): Both of order $O(\bar{f})$ under (A3a)–(A3c) — uniformly small relative to the direct effect. Hence $dG/dy < 0$. □

Proof: Single Crossing Lemma ($\tilde{y}_G < \tilde{y}_B$)

Recall $\Delta V(y, \theta) = \pi_\theta G(y) + (1 - \pi_\theta)\Delta \underline{u}$.

Part (i): ΔV strictly decreasing in y for $y > y^{}$**

$$\frac{d\Delta V}{dy} = \pi_\theta \frac{dG}{dy} < 0$$

because $dG/dy < 0$ (Wealth Effect Lemma) and $\pi_\theta > 0$.

Part (ii): Good types exit strictly before Bad types

Each type's threshold $\tilde{y}_\theta$ is unique (strict monotonicity). At the Good type's exit income $\tilde{y}_G$, evaluate the Bad type's gain:

$$\Delta V(\tilde{y}_G, B) = \pi_B G(\tilde{y}_G) + (1 - \pi_B)\Delta \underline{u} = \frac{\Delta \pi \Delta \underline{u}}{\pi_G} > 0,$$

using $G(\tilde{y}_G) = -\frac{1-\pi_G}{\pi_G} \Delta \underline{u}$ and $\Delta \pi = \pi_G - \pi_B > 0$.

The Bad type *still prefers cosigning* at the Good type's exit. Since $\Delta V(\cdot, B)$ is strictly decreasing, the Bad type's threshold satisfies $\tilde{y}_B > \tilde{y}_G$. $\square$